A Project Report on

LOST AND FOUND PORTAL

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

By

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A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

2025-2026

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CERTIFICATE

This is to certify that the project entitled "Lost and Found Portal" is a bonafide work carried out by Vedant Satish Mayeakar (24106085), Prathmesh Milind More (24106196), Manthan Dinesh Patil (24106011), and Yatharth Mahesh Patne (24106132), submitted to the University of Mumbai in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning).

Dr. Rahul Ambekar	Prof. Vijaya Bharti
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Project Report Approval for S.E.

This project report entitled "Lost and Found Portal" by Vedant Satish Mayekar, Prathmesh Milind More, Manthan Dinesh Patil, and Yatharth Mahesh Patne is approved for the award of the degree of ***Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning) for the academic year 2025–26.***

Examiner Name	Signature
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2._____	_____

Date:

Place:

Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

In many educational institutions, the management of lost and found items has traditionally relied on manual methods such as registers and notice boards, often resulting in inefficiencies, delays, and a lack of proper tracking. To overcome these limitations, the “Institute Level Lost and Found Portal with Senior-to-Junior Academic Exchange System” was developed as a centralized web-based solution. The platform replaces conventional practices with a structured and transparent digital system, enabling users to easily report, search, and manage lost or found items from a single interface.

The portal incorporates essential features such as image-based item reporting, detailed item descriptions, and real-time status tracking, which significantly improve the chances of recovering lost belongings. It also includes role-based authentication and a systematic claim verification workflow to ensure data security, authenticity, and accountability. These functionalities not only enhance the reliability of the system but also reduce the possibility of errors, duplication, and misuse.

Beyond its primary functionality, the system introduces a Senior-to-Junior Academic Exchange module aimed at promoting collaboration and resource optimization within the institution. Through this feature, senior students can sell, donate, or share academic materials such as textbooks, lab instruments, and accessories with junior students. This initiative helps in reducing financial burden, minimizing resource wastage, and fostering a culture of sharing and mutual support among students.

Additionally, the platform provides analytical insights such as recovery rates, user engagement, and usage trends, enabling administrators to monitor system performance and make data-driven decisions. The user-friendly interface and efficient design ensure accessibility and ease of use for all stakeholders. Overall, the project represents a significant step towards digital transformation, improving institutional efficiency while creating a more connected, sustainable, and resource-efficient academic environment.

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ABBREVAIONS

AI	*Artificial Intelligence*
ML	*Machine Learning*
DBMS	*Database Management System*
UI	*User Interface*
UX	*User Experience*
API	*Application Programming Interface*
HTML	*HyperText Markup Language*
CSS	*Cascading Style Sheets*
JSP	*Java Server Pages*
SQL	*Structured Query Language*
ERP	*Enterprise Resource Planning*
HTTP	*HyperText Transfer Protocol*

Chapter 1

Introduction

1.1 Overview

In educational institutions, the issue of lost and misplaced items is common and unavoidable. Students and faculty members frequently lose personal belongings such as identity cards, wallets, mobile phones, books, laboratory equipment, and important documents. These items are essential for daily academic activities, and their loss can cause inconvenience and disruption.

Traditional methods used to manage lost and found items include manual registers maintained at security desks, notice boards, or informal communication through messaging groups. However, these approaches are inefficient, time-consuming, and lack a structured system for tracking and verification. There is no proper way to maintain records, verify ownership, or monitor claim status, which often leads to delays or unsuccessful recovery of items.

To overcome these limitations, the “Institute Level Lost and Found Portal with Senior-to-Junior Academic Exchange System” is proposed as a web-based solution. The system provides a centralized digital platform where users can report lost or found items, upload images, search for items, and submit claims. By digitizing the entire process, the system improves efficiency, transparency, and reliability.

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Abstract

In many educational institutions, the management of lost and found items has traditionally relied on manual methods such as registers and notice boards, often resulting in inefficiencies, delays, and lack of proper tracking. To overcome these limitations, the proposed *Institute Level Lost and Found Portal with Senior-to-Junior Academic Exchange System* is developed as a centralized web-based solution. The platform replaces conventional practices with a structured and transparent digital system, enabling users to efficiently report, search, and manage lost or found items through a single interface.

Beyond its primary functionality, the platform introduces a Senior-to-Junior Academic Exchange module aimed at promoting collaboration and resource optimization within the institution. Through this feature, senior students can sell, donate, or share academic materials such as textbooks, laboratory instruments, and accessories with junior students. This initiative helps in reducing financial burden, minimizing resource wastage, and fostering a culture of sharing and mutual support.

The project aligns with several Sustainable Development Goals (SDGs) defined by the United Nations under the 2030 Agenda for Sustainable Development. It contributes to **SDG 4: Quality Education** by improving access to learning resources, **SDG 9: Industry, Innovation and Infrastructure** through the implementation of a digital platform, **SDG 10: Reduced Inequalities** by supporting resource sharing among students, **SDG 11: Sustainable Cities and Communities** by enhancing institutional efficiency, **SDG 12: Responsible Consumption and Production** by encouraging reuse of materials, and **SDG 17: Partnerships for the Goals** by promoting collaboration between students.

Additionally, the platform provides analytical insights such as recovery rates, user engagement, and usage trends, enabling administrators to monitor system performance and make data-driven decisions. The user-friendly interface and efficient design ensure accessibility and ease of use for all stakeholders. Overall, the project represents a significant step toward digital transformation, improving institutional efficiency while creating a more connected, sustainable, and resource-efficient academic environment.

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Chapter 1

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To overcome these limitations, the “Institute Level Lost and Found Portal with Senior-to-Junior Academic Exchange System” is proposed as a web-based solution. The system provides a centralized digital platform where users can report lost or found items, upload images, search for items, and submit claims. By digitizing the entire process, the system improves efficiency, transparency, and reliability.

1.2 Need for the System

The need for this system arises from the shortcomings of traditional manual methods. Manual record-keeping is prone to human errors, duplication, and data loss. It is also difficult to retrieve past records or track item status effectively. Additionally, there is no proper verification mechanism, which increases the risk of false claims.

Another important issue is the lack of integration between different functionalities. While some platforms provide lost-and-found services, others focus on academic material exchange, but very few systems combine both features into a single platform. This creates inefficiency and limits usability.

With the increasing adoption of digital technologies in educational institutions, there is a strong need for a smart and automated system that simplifies operations, improves accessibility, and enhances user experience.

1.3 Senior-to-Junior Academic Exchange

An important feature of the proposed system is the Senior-to-Junior Academic Exchange module. Senior students often possess academic materials such as textbooks, calculators, and lab equipment that are no longer required after course completion. These items are usually unused or wasted.

At the same time, junior students often face financial constraints in purchasing new academic materials. The exchange system addresses this problem by allowing senior students to upload details of items they wish to sell or donate. Junior students can browse these items and acquire them at lower cost or free of charge.

This feature promotes resource sharing, reduces wastage, and creates a collaborative environment within the institution. It also enhances the overall usefulness of the system beyond lost-and-found management.

1.4 Key Features and Advantages

The proposed system provides a centralized platform for managing lost and found items digitally. Users can report items with descriptions and images, making identification easier. The system includes search functionality and a structured claim verification process, ensuring transparency and accountability.

The academic exchange module enables efficient sharing of resources among students. Additionally, the system provides basic analytics such as recovery statistics, helping administrators monitor performance.

The digital approach offers several advantages, including reduced manual work, improved efficiency, secure data handling, and real-time access to information. It also supports the concept of a smart campus by integrating multiple functionalities into a single.

Chapter 2

Literature Survey

2.1 Introduction

With the advancement of digital technologies, educational institutions are increasingly adopting web-based systems to improve administrative efficiency and user experience. Lost-and-found management and academic resource sharing are two important areas where digital solutions can significantly enhance operations. Various systems have been developed to address these problems individually; however, most of them lack integration, structured workflows, and proper security mechanisms.

This literature review examines recent systems (2021–2025) related to lost-and-found management and academic exchange platforms, identifies their limitations, and highlights the research gap addressed by the proposed system.

2.2 Review of Recent Research (2021–2025)

The following table summarizes recent research papers and systems related to lost-and-found and academic exchange platforms:

Table 1: ***Existing system***

Sr. No.	Author(s)	Year	Title	Key Contribution	Limitation
1	S. Sharma et al.	2022	Smart Lost Item Tracking System	Introduced notification-based tracking system	No claim verification process
2	R. Patel & M. Mehta	2022	Academic Resource Sharing Platform	Enabled peer-to-peer sharing of study materials	Not integrated with institutions
3	A. Kumar & N. Singh	2022	Web-Based Lost and Found System	Centralized reporting and searching of items	Limited security features
4	D. Mishra et al.	2023	Smart Campus Lost & Found System	Improved item categorization and search filters	No analytics module
5	J. Lee & K. Park	2023	QR Code-Based Item Tracking	Faster item identification using QR codes	Requires additional infrastructure
6	V. Verma & R. Singh	2023	Centralized Lost and Found Portal	Structured database for item management	No lifecycle tracking
7	P. Nair & S. Menon	2024	Web-Based Campus Management System	Integrated campus services into one platform	Lacks exchange module
8	A. Gupta et al.	2024	AI-Based Object Matching System	Image recognition for item matching	High complexity and cost
9	K. Desai & H. Shah	2025	Student Resource Sharing System	Efficient academic exchange mechanism	No verification workflow
10	S. Jain & R. Agarwal	2025	Smart Inventory & Lost Item System	Combined inventory tracking with lost item management	Limited user interface design

2.3 Existing Lost and Found Systems

Several web-based lost-and-found systems have been developed to replace traditional manual methods. These systems typically allow users to report lost or found items by entering details such as item description, location, and time. Some platforms also provide search functionality to help users find matching items.While these systems improve accessibility and reduce manual effort, they often lack structured verification mechanisms. Most systems do not include an admin approval workflow, which increases the risk of false claims. Additionally, many platforms do not maintain a complete lifecycle of items, such as tracking status from "lost" to "claimed" and "recovered."Recent systems (2021–2025) have introduced features such as notifications and QR-based tracking, but they still face challenges related to cost, complexity, and lack of integration.

2.4 Academic Exchange Platforms

Academic exchange platforms are designed to facilitate the sharing or selling of educational materials among students. These platforms allow users to list items such as textbooks, notes, and equipment, which can be accessed by other students.

Although these systems promote resource sharing and reduce costs, they are usually developed as independent marketplaces. They are not integrated with institutional systems, which leads to issues such as lack of authentication, security concerns, and absence of administrative control.Recent research (2023 onwards) highlights peer-to-peer exchange systems, but these platforms still lack proper monitoring, verification, and institutional support.

2.5 Limitations of Existing Systems

Despite the advancements in recent lost-and-found and academic exchange systems, several limitations still exist that reduce their overall effectiveness. One of the major issues is the lack of integration between different functionalities. Most platforms focus either on lost-and-found management or academic resource exchange, but they do not combine both features into a single system, leading to inefficiency and dependence on multiple platforms.

Another key limitation is the absence of a structured verification workflow. Many systems allow users to claim items without proper authentication or administrative approval, increasing the risk of false claims. Additionally, these systems often fail to maintain a complete lifecycle of items, making it difficult to track their status from reporting to recovery.

Furthermore, most systems do not provide analytical features such as recovery rate or performance monitoring. Security is also a concern due to weak authentication mechanisms. Although advanced solutions like AI and QR-based systems exist, they are often costly and complex, making them less practical for institutional use.

2.6 Research Gap

The analysis of recent research (2021–2025) reveals a clear gap in the development of a unified system that integrates both lost-and-found management and academic exchange. Existing solutions are mostly independent and lack comprehensive functionality, limiting their effectiveness in real-world institutional environments.

Another important gap is the lack of structured claim verification and lifecycle tracking. Many systems do not include proper administrative control, which reduces reliability and trust. Additionally, analytical features such as recovery statistics and performance evaluation are often missing, making it difficult to assess system efficiency.

There is also a need for better security through role-based access control and authentication mechanisms. Moreover, most systems fail to provide a cost-effective and scalable solution suitable for educational institutions. The proposed system addresses these gaps by offering an integrated, secure, and efficient platform with both lost-and-found and Senior-to-Junior exchange functionalities.

In addition to these limitations, existing systems often lack user-centric design and ease of use, which affects their adoption among students and faculty members. Many platforms are either too complex or not tailored to institutional needs, resulting in low engagement. Therefore, there is a strong requirement for a system that not only integrates multiple functionalities but also provides a simple, intuitive, and user-friendly interface. The proposed system aims to fulfill this requirement by ensuring accessibility, ease of navigation, and practical usability within an academic environment

Chapter 3

Limitation of Existing system

3.1 Introduction

Despite the availability of various methods for managing lost and found items, most existing systems still face several challenges that reduce their efficiency and reliability. Many institutions continue to depend on traditional or partially digital systems, which are not capable of handling large amounts of data effectively. These limitations lead to delays in item recovery, lack of transparency, and poor user experience.

In addition, current systems do not fully utilize modern technologies to provide structured workflows, secure verification, and proper tracking mechanisms. This chapter highlights the major limitations of existing systems and explains why there is a need for a more advanced and integrated solution.

3.2 Manual Record-Keeping Issues

One of the most widely used methods in institutions is maintaining physical registers at security desks. Although this method is simple and inexpensive, it suffers from several drawbacks. Recording item details manually often leads to incomplete or unclear information, which makes it difficult to identify items correctly.

Searching for a specific item in a register is time-consuming, especially when records increase over time. There is no proper categorization or indexing, which makes retrieval inefficient. Moreover, registers can be easily damaged, lost, or misplaced, resulting in permanent loss of

data. Another issue with manual systems is the lack of real-time updates. Users cannot access information instantly and must rely on physical visits or inquiries. This significantly reduces the speed and effectiveness of the entire process.

3.3 Lack of Centralized Platform

Existing systems often lack a centralized platform for managing lost and found data. Information is usually scattered across multiple sources such as notice boards, security offices, and messaging groups. This fragmentation creates confusion and makes it difficult for users to access accurate and complete information.

Without a centralized system, there is no uniform method for reporting, searching, or claiming items. Different departments may follow different procedures, leading to inconsistency in data handling. Users are forced to check multiple sources, which increases effort and reduces efficiency. A centralized platform is essential to ensure that all information is stored in one place, making it easily accessible and manageable for both users and administrators.

3.4 Absence of Verification Mechanism

A major limitation of many existing systems is the lack of a proper verification process for claim approval. In some cases, items are returned without confirming the identity of the claimant, which increases the risk of false claims.

There is no structured workflow that ensures proper validation of ownership, such as checking item details, proof, or supporting information. This lack of verification not only leads to misuse but also reduces trust in the system. Furthermore, there is no proper record of claim approvals or rejections, making it difficult to maintain accountability. A well-defined verification mechanism is necessary to ensure fairness and security in the system.

3.5 Limited Tracking and Monitoring

Another important limitation is the lack of tracking and monitoring capabilities. Most systems do not maintain a complete record of an item's journey from the time it is reported lost to when it is found and finally returned to the owner.

Without proper tracking, it becomes difficult to determine the status of items or measure the success rate of recovery. Users are often unaware of whether their item has been found or claimed. Additionally, existing systems do not provide analytical features such as recovery rate, monthly trends, or performance reports. These insights are important for administrators to evaluate system efficiency and identify areas for improvement.

3.6 Security and Authentication Issues

Security and authentication are critical aspects of any system that handles user data and sensitive information. In many existing lost-and-found systems, security is often overlooked or implemented in a basic manner, which leads to several vulnerabilities. These systems usually

do not have strong authentication mechanisms, allowing unauthorized users to access the platform. As a result, user data such as personal details, item information, and claim records may be exposed or misused. This lack of security not only affects system reliability but also reduces user trust.

One of the major issues in existing systems is the absence of proper authentication methods. Many platforms allow users to access functionalities without verifying their identity through secure login processes. Even when login systems are present, they often lack strong password policies, encryption techniques, or multi-level authentication. This makes it easier for attackers to gain unauthorized access through simple methods such as password guessing or credential theft. Without secure authentication, it becomes difficult to ensure that only legitimate users are interacting with the system.

Another important limitation is the lack of role-based access control. In many systems, all users are given similar access privileges, which can lead to misuse of system functionalities. For example, a general user may be able to modify or delete item records, which should ideally be restricted to administrators. This absence of controlled access increases the risk of data manipulation and reduces system integrity. A well-designed system should clearly define roles such as admin and user, and assign permissions accordingly to maintain proper control.

Data protection is also a major concern in existing systems. Sensitive data is often stored without proper encryption, making it vulnerable to data breaches. Additionally, there is little or no validation of user input, which can lead to security threats such as injection attacks or malicious data entry. Without proper validation and sanitization techniques, the system becomes susceptible to various forms of cyber attacks.

Furthermore, many systems do not maintain proper logs or records of user activities. This makes it difficult to track unauthorized actions or identify security breaches. The absence of monitoring mechanisms reduces accountability and makes it challenging to investigate issues when they arise.

Therefore, it is essential to implement strong security measures in any modern system. These include secure authentication mechanisms, role-based access control, data encryption, input validation, and activity logging. By addressing these issues, the proposed system ensures better protection of user data, enhances system reliability, and builds trust among users.

Chapter 4

Problem Statement and Objectives

4.1 Problem Statement

In educational institutions, the management of lost and found items is a persistent problem due to the absence of a structured and centralized system. Students and faculty members frequently lose important belongings such as identity cards, wallets, mobile phones, books, laboratory equipment, and documents that are essential for academic activities. The current methods used to handle such situations are mostly manual, including maintaining registers at security desks, displaying notices on bulletin boards, or sharing information through informal communication channels like messaging groups. These methods are inefficient, time-consuming, and lack proper organization, making it difficult to track and recover lost items effectively.

Manual systems are highly prone to errors, including incomplete data entry, duplication of records, and even loss of information over time. Searching for a specific item in large registers is difficult and often leads to delays. Additionally, there is no proper verification mechanism to confirm ownership when an item is claimed, which increases the chances of false claims and misuse. Another major issue is the lack of item lifecycle tracking, meaning there is no clear record of whether an item is lost, found, claimed, or returned. This reduces transparency and makes it difficult for administrators to monitor system performance or ensure accountability.

Apart from lost-and-found management, another important problem exists within academic institutions. Senior students often possess academic materials such as textbooks, calculators, and laboratory accessories that are no longer needed after course completion. These items are frequently unused or discarded, leading to wastage of valuable resources. At the same time, junior students face financial challenges in purchasing new materials. However, there is no structured institutional platform that connects seniors and juniors for sharing or selling such resources. This lack of integration results in missed opportunities for cost savings and efficient resource utilization.

Therefore, there is a strong need for a centralized, secure, and efficient system that can address both problems simultaneously. The system should provide proper tracking, verification, and management of lost items while also enabling a structured platform for academic exchange. Such a system would improve efficiency, reduce manual effort, enhance transparency, and promote sustainable use of resources within the institution.

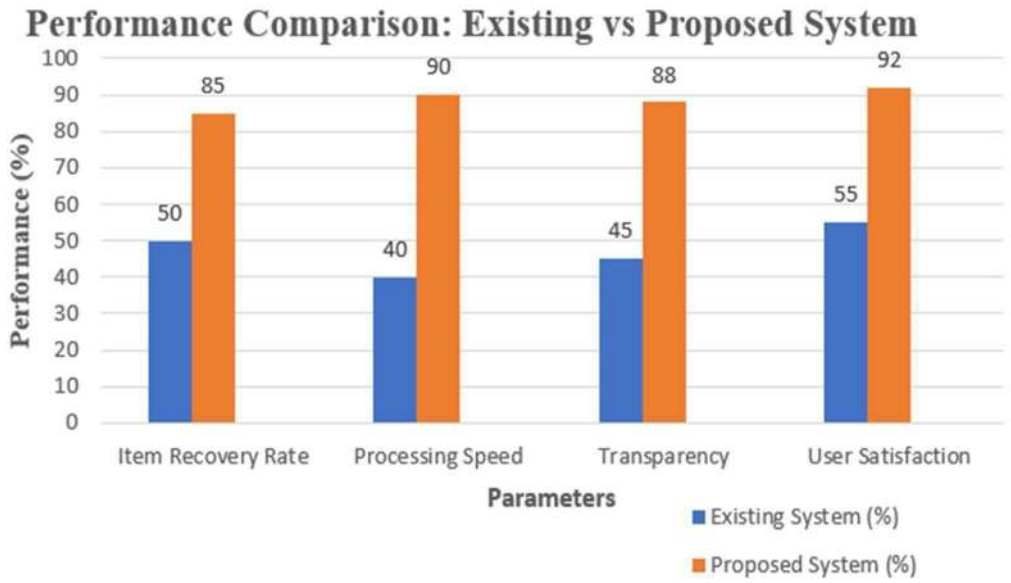
4.2 Objectives of the System

The main objective of this project is to develop a web-based platform that efficiently manages lost and found items while also facilitating academic resource exchange between students in an organized and secure manner. The system aims to replace traditional manual methods with a digital solution that ensures better accessibility, faster processing, and improved accuracy in managing records. By integrating multiple functionalities into a single platform, the system enhances overall efficiency and user experience within the institution.

To achieve this objective, the system focuses on the following key aspects:

- **Centralized Platform:** The system provides a unified platform where users can report lost or found items with detailed descriptions and images. This helps in accurate identification and improves the chances of successful recovery while eliminating the need for manual registers**.**
- **Efficient Search Mechanism:** A powerful search functionality is implemented to allow users to quickly locate items using keywords, categories, or filters. This reduces the time required to find lost items and improves system usability.
- **Secure Authentication and Access Control:** The system includes a secure login mechanism with role-based access control, ensuring that only authorized users can access specific features. This enhances system security and prevents unauthorized actions.
- **Structured Claim Verification:** A proper claim approval process is implemented where administrators verify and approve requests based on valid information. This ensures authenticity and reduces the chances of false claims.
- **Senior-to-Junior Academic Exchange Module:** The system introduces a feature that allows senior students to upload academic materials for sale or donation, while junior students can easily access these resources. This promotes resource sharing, reduces financial burden, and minimizes wastage.
- **Analytical and Performance MonitoringFeatures:** The system provides insights such as recovery rate, usage statistics, and system performance analysis. These features help administrators evaluate efficiency and make informed decisions for improvement.

4.3 System Performance Graph



GRAPH 1: **Performance Comparison**

4.4 Interpretation

The graph clearly illustrates a significant improvement in overall system performance after the implementation of the proposed solution. One of the most noticeable enhancements is in the item recovery rate, which increases due to the use of a centralized database and efficient search functionality. Unlike traditional systems where users depend on manual checking, the proposed system allows quick identification of items through keywords, categories, and images. This structured approach improves accuracy and increases the chances of successfully recovering lost items.

Another important improvement observed is in processing speed. In existing systems, tasks such as recording item details, searching records, and verifying claims are performed manually, which consumes a considerable amount of time. The proposed system automates these processes, enabling real-time data entry, faster search results, and quick claim handling. This reduction in manual effort significantly enhances system efficiency and ensures that users can complete tasks without delays.

Furthermore, the system shows a considerable increase in transparency and user satisfaction. The introduction of structured workflows and a proper claim verification mechanism ensures accountability and prevents misuse. Users can track the status of their items, which builds trust in the system. Additionally, the inclusion of features such as the Senior-to-Junior academic exchange module improves usability and provides added value. The combination of ease of use, faster processing, and enhanced functionality leads to higher user satisfaction and better overall system performance.

Chapter 5

Proposed System

5.1 Overview

The proposed system is a web-based application designed to provide a centralized and efficient solution for managing lost and found items within an educational institution. It replaces traditional manual methods with a digital platform that ensures better accessibility, faster processing, and improved transparency. Users can report lost or found items by entering relevant details such as item name, description, location, and images. This structured data helps in accurate identification and improves the chances of successful recovery.

In addition to basic functionality, the system introduces an integrated approach by combining lost-and-found management with a Senior-to-Junior academic exchange module. This feature allows students to reuse academic materials, thereby reducing costs and promoting sustainability. The system is designed to be user-friendly, ensuring that even non-technical users can easily navigate and utilize its features effectively.

5.2 Advantages of Proposed System

The proposed system offers several advantages over traditional methods by providing a centralized and automated platform. It reduces manual work, minimizes errors, and improves efficiency in managing lost-and-found operations. The structured workflow ensures transparency and accountability, making the system reliable and trustworthy.

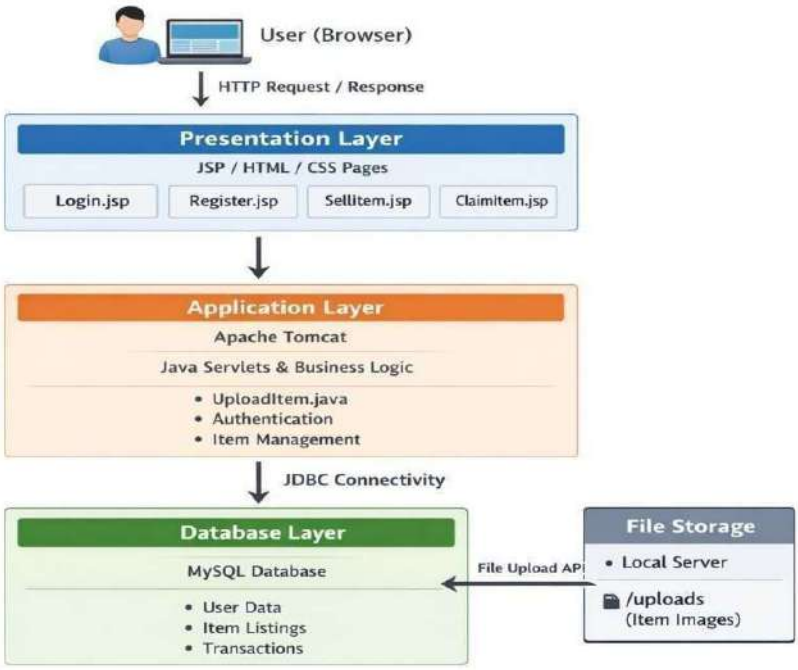
Additionally, the integration of the academic exchange module enhances the system's usefulness by promoting resource sharing. The system also provides better data management,

faster processing, and improved user experience. Its scalable design allows future enhancements, making it suitable for long-term institutional use.

5.3 System Architecture

The system follows a three-tier architecture, which ensures separation of concerns and enhances scalability, maintainability, and performance. The presentation layer provides an interactive interface where users can perform operations such as reporting items, searching records, and submitting claims. It is designed using modern front-end technologies to ensure responsiveness and ease of use.

The application layer handles all business logic, including authentication, item management, claim verification, and exchange operations. It acts as a bridge between the user interface and the database, ensuring smooth data flow and processing. The database layer stores all system data in a structured format, including user details, item records, and transaction history. Proper database design techniques such as normalization and indexing are used to ensure efficient data retrieval and integrity



5.4 Workflow of the System

The workflow of the system is designed to ensure smooth and efficient handling of lost-and-found operations. Initially, users log into the system and report items by providing necessary details. The system stores this information in the database and makes it available for search operations. Other users can browse or search for items using keywords, which simplifies the process of locating lost belongings.

When a user identifies a matching item, they can submit a claim request. The request is forwarded to the administrator, who verifies the details and approves or rejects the claim. Once approved, the item is returned to the rightful owner, and the system updates its status. Similarly, in the academic exchange module, seniors upload items, and juniors can browse and acquire them, ensuring efficient resource utilization.

5.5 Security and Data Management

The proposed system places strong emphasis on security and efficient data management to ensure reliability and user trust. Since the system handles sensitive information such as user details, item records, and claim requests, it is essential to implement proper security mechanisms. The system incorporates user authentication, where each user must log in using valid credentials before accessing the platform. This prevents unauthorized access and ensures that only registered users can perform operations such as reporting items or submitting claims.

In addition to authentication, role-based access control is implemented to differentiate between users and administrators. While general users can report and search items, administrative privileges are restricted to authorized personnel who can verify claims and manage records. This structured access control enhances system security and prevents misuse of functionalities. Furthermore, input validation techniques are applied to ensure that only valid and accurate data is entered into the system, reducing the chances of errors and malicious inputs.

From a data management perspective, the system uses a structured database to store and organize information efficiently. Techniques such as data normalization are applied to avoid redundancy and maintain consistency. Regular updates and proper indexing ensure faster data retrieval, which improves overall system performance. Backup mechanisms can also be implemented to prevent data loss in case of system failures. Overall, the combination of security measures and efficient data management ensures that the system remains reliable, secure, and scalable for long-term use.

Chapter 6

Experimental Setup

6.1 Introduction

The experimental setup describes the environment, tools, and procedures used for the development, testing, and validation of the proposed system. It plays a crucial role in ensuring that the system functions correctly under different conditions and meets all specified requirements. A well-defined experimental setup helps in evaluating system performance, identifying errors, and ensuring reliability before deployment. In this project, both hardware and software components are carefully selected to provide a stable and efficient development environment.

The setup is designed to support all aspects of the system, including front-end development, back-end processing, and database management. It ensures smooth interaction between different system components and allows testing under real-time conditions. Proper configuration of tools and technologies also helps in improving system performance, minimizing errors, and ensuring that the system behaves consistently across different environments.

Key objectives of the experimental setup include:

- To provide a stable development environment
- To ensure proper integration of system components
- To evaluate system performance under different conditions

6.2 Hardware and Software Environment

The system is developed using a standard computing environment that is capable of handling web-based applications efficiently. A computer system with a minimum of 8GB RAM and a Windows operating system is used to ensure smooth execution of development tools, servers, and databases. This configuration supports multitasking, allowing simultaneous execution of front-end, back-end, and database operations without performance degradation. The hardware setup ensures that the system can handle multiple user interactions and data processing tasks effectively.

On the software side, various modern technologies are used to build the system. The front-end is developed using HTML, CSS, and React.js, which provide an interactive and responsive user interface. These technologies ensure that the system is visually appealing, user-friendly, and compatible with different devices. The back-end is implemented using Java and JSP, which handle business logic, user authentication, and communication with the database efficiently.

MySQL is used as the database management system to store structured data such as user details, item records, claims, and exchange information. Apache Tomcat is used as the application server for deploying and running the system. Visual Studio Code serves as the integrated development environment (IDE), providing features such as code editing, debugging, and version control support.

6.2.1 Hardware Components:

- Processor: Intel i5 or above
- RAM: Minimum 8GB
- Storage: 256GB or higher
- Operating System: Windows 10/11

6.2.2 Software Components:

- Front-end: HTML, CSS,
- Back-end: Java, JSP
- Database: MySQL
- Server: Apache Tomcat
- IDE: Visual Studio Code

This combination of hardware and software ensures efficient system development, smooth execution, and reliable performance.

6.3 Deployment and Testing Setup

The system is deployed locally using the Apache Tomcat server, which allows it to be accessed through a web browser. The database is configured using MySQL Workbench, and proper connections are established between the application and database. This setup ensures that data can be stored, retrieved, and updated efficiently. The deployment environment is designed to simulate real-world usage conditions, allowing accurate evaluation of system performance.

Testing is carried out to evaluate the functionality, performance, and security of the system. Functional testing ensures that all modules operate correctly and meet user requirements. Each module is tested individually and then integrated to verify overall system functionality. Performance testing is conducted to measure system response time and ensure that it can handle multiple user requests efficiently without delays.

Security testing is also performed to verify that the system implements proper authentication and role-based access control. This ensures that only authorized users can access specific features and prevents unauthorized access to sensitive data. Additionally, stress testing is conducted to evaluate system stability under high load conditions, ensuring that the system remains stable and responsive even when multiple users access it simultaneously.

Types of Testing Performed:

- Functional Testing: Verifies correctness of system modules
- Performance Testing: Measures speed and response time
- Security Testing: Ensures safe data handling and access control
- Stress Testing: Evaluates system stability under heavy load

The results of testing indicate that the system performs efficiently and meets all functional and non-functional requirements. The system is stable, secure, and capable of handling real-time operations, making it suitable for deployment in educational institutions.

Chapter 7

Results and Discussion

7.1 Improvement in Item Recovery Rate

The proposed system shows a significant improvement in the recovery rate of lost items compared to traditional manual methods. In existing systems, the recovery process largely depends on chance, physical checking of registers, or informal communication. This often leads to delays and reduces the probability of successfully finding lost items. However, the proposed system introduces a centralized digital platform where all item records are stored systematically, making it easier to locate items.

The implementation of advanced search functionality plays a key role in improving recovery rates. Users can search for items using keywords, descriptions, or categories, which allows them to quickly identify potential matches. The addition of image upload further enhances the accuracy of identification, as users can visually confirm whether an item belongs to them. This reduces confusion and speeds up the recovery process.

Furthermore, the system maintains proper tracking of item status, such as lost, found, claimed, and recovered. This lifecycle tracking ensures that users are always aware of the current status of their items. As a result, the overall efficiency of the recovery process improves, leading to higher success rates and better user satisfaction.

7.2 Efficient Claim Management and Verification

The claim management process in the proposed system is designed to ensure accuracy, transparency, and security. In traditional systems, there is often no proper verification mechanism, and items may be handed over without confirming ownership. This increases the

risk of false claims and misuse. The proposed system addresses this issue by introducing a structured claim verification workflow.

When a user identifies a matching item, they can submit a claim request through the system. The request is then reviewed by the administrator, who verifies the details provided by the user. This may include checking item descriptions, supporting information, or other relevant data. Only after proper verification is the claim approved, ensuring that the item is returned to its rightful owner.

This structured approach improves accountability and builds trust among users. The system also maintains records of all claims, including approvals and rejections, which can be used for future reference. This not only enhances transparency but also helps in monitoring system performance and identifying potential issues.

7.3 User Experience and System Performance

The proposed system significantly enhances user experience by providing a simple, intuitive, and efficient interface. Unlike traditional methods that require physical effort and time, the digital platform allows users to perform all operations online. Reporting items, searching records, and submitting claims can be done quickly and easily, making the system more convenient for users.

The system also demonstrates high performance in terms of processing speed and responsiveness. Since most operations are automated, tasks are completed in real time without delays. The use of modern web technologies ensures that the system is responsive across different devices, allowing users to access it from desktops, laptops, or mobile devices.

Additionally, the inclusion of the Senior-to-Junior academic exchange module further enhances the system's usability. This feature allows students to share or sell academic materials, creating additional value beyond lost-and-found management. Overall, the system provides a smooth and efficient user experience, which contributes to higher user satisfaction and increased adoption.

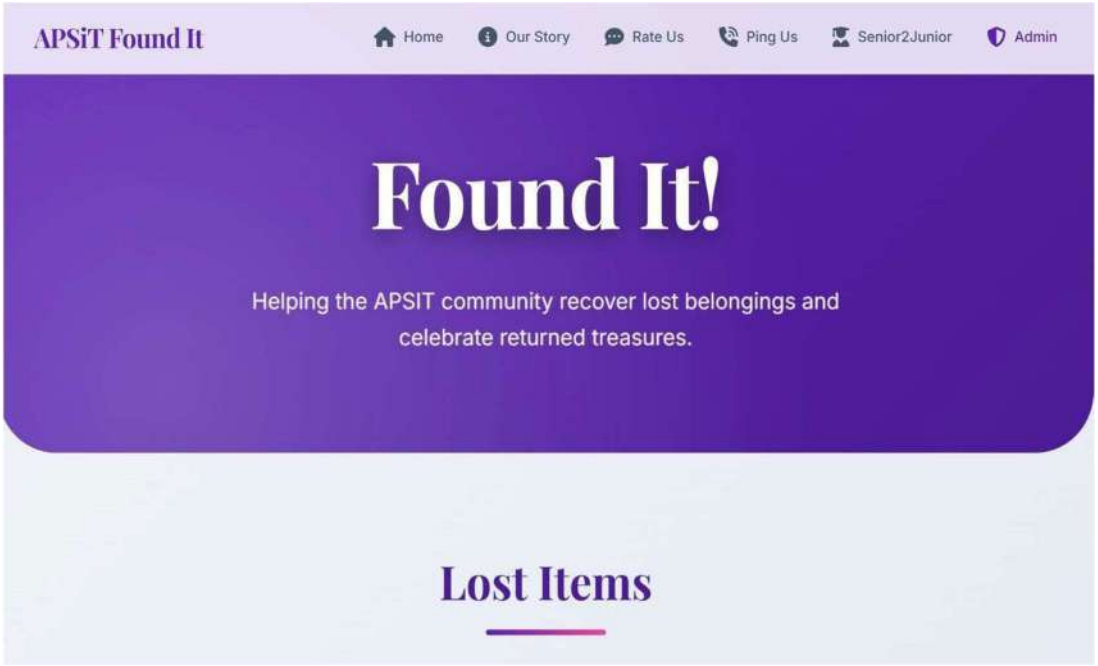


FIGURE 1: **Home Page**

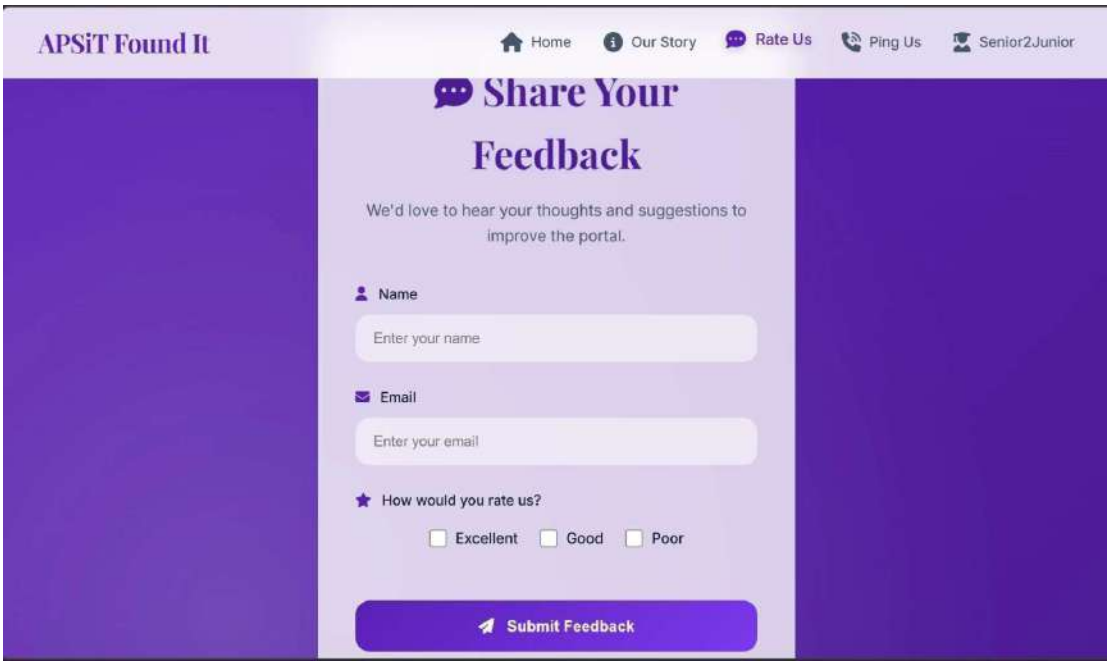


FIGURE 2: **Feedback Page**

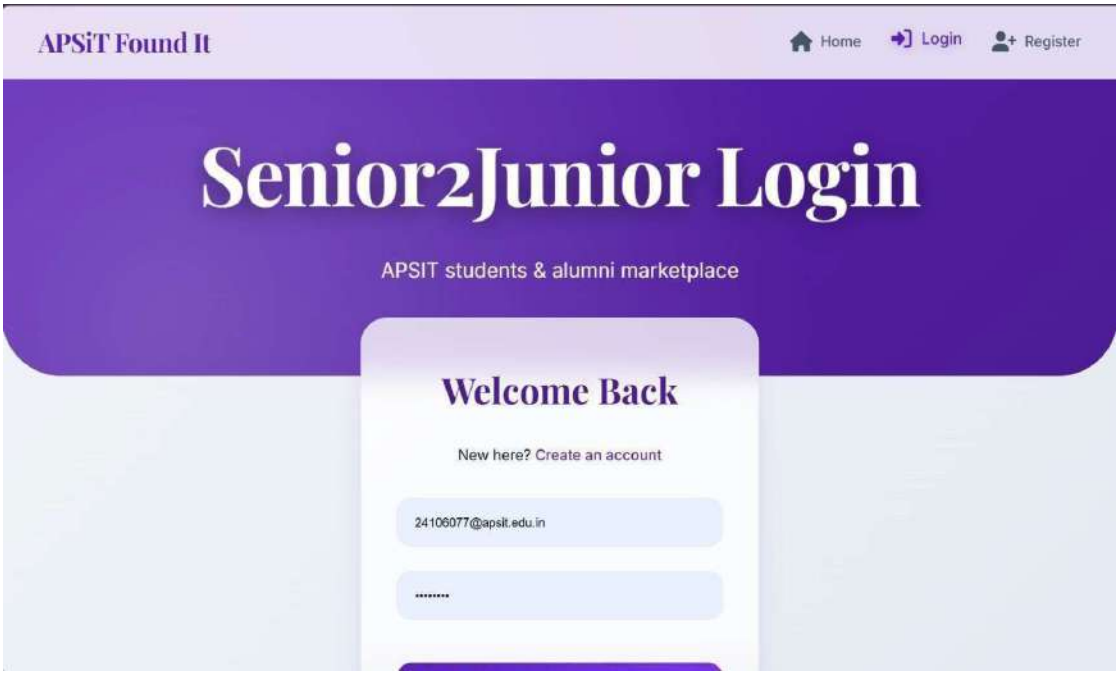


FIGURE 3: **Senior2Junior**

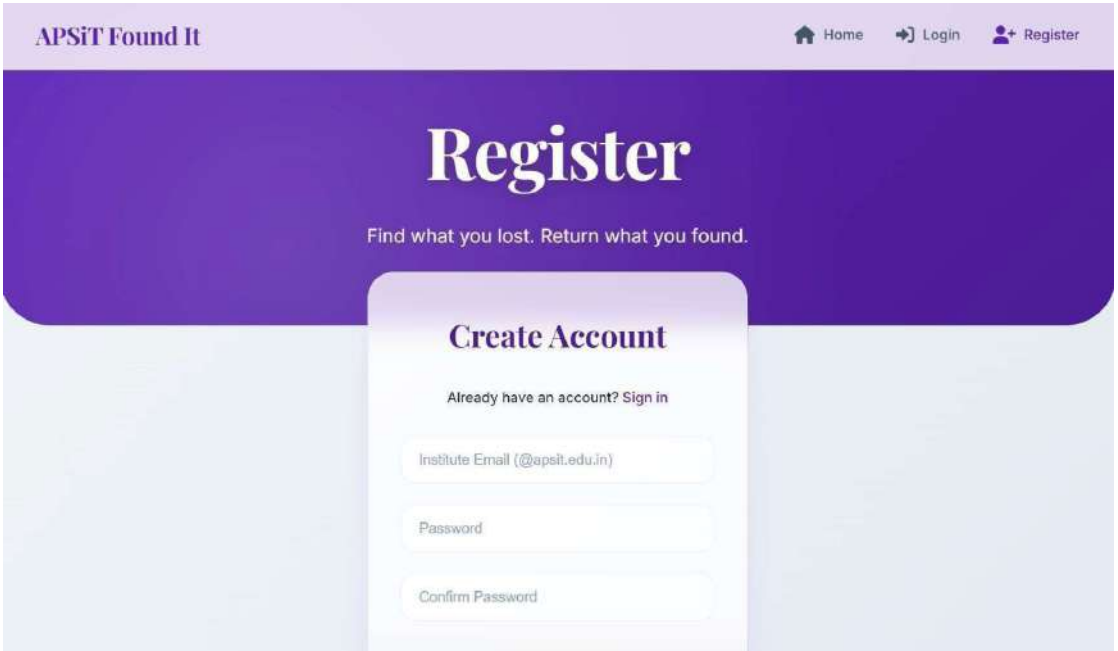


FIGURE 4: **Register Page**

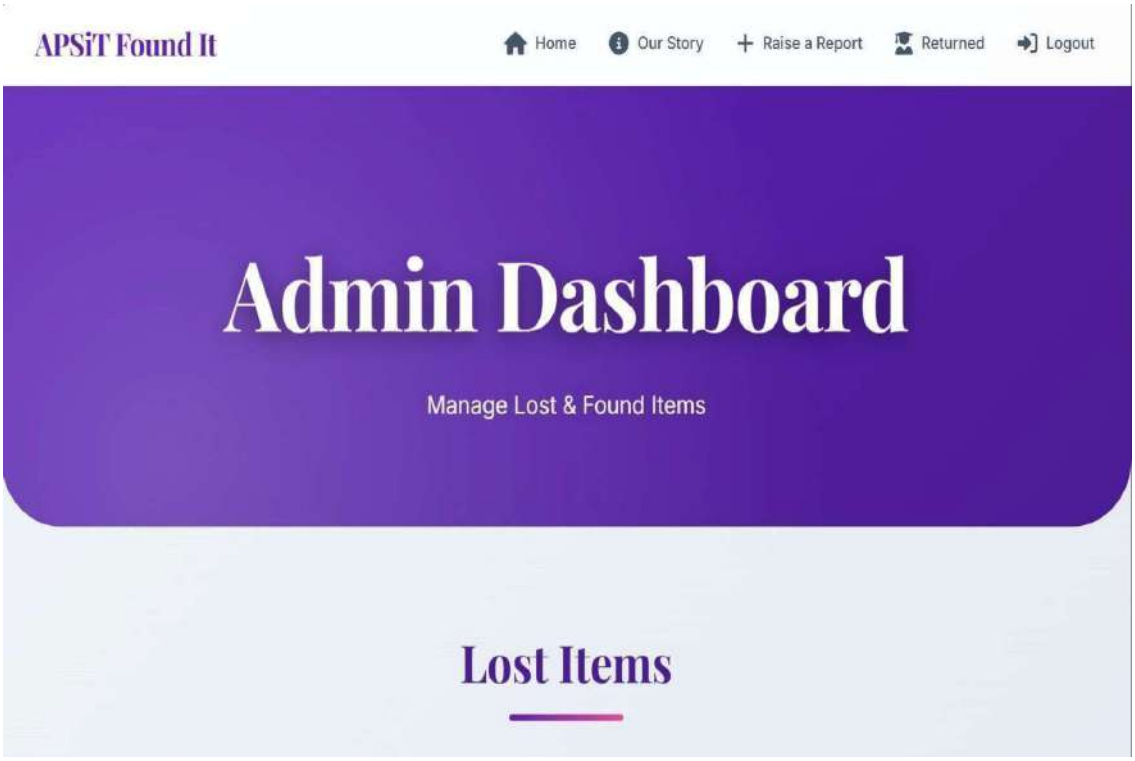


FIGURE 5: **Admin Page**

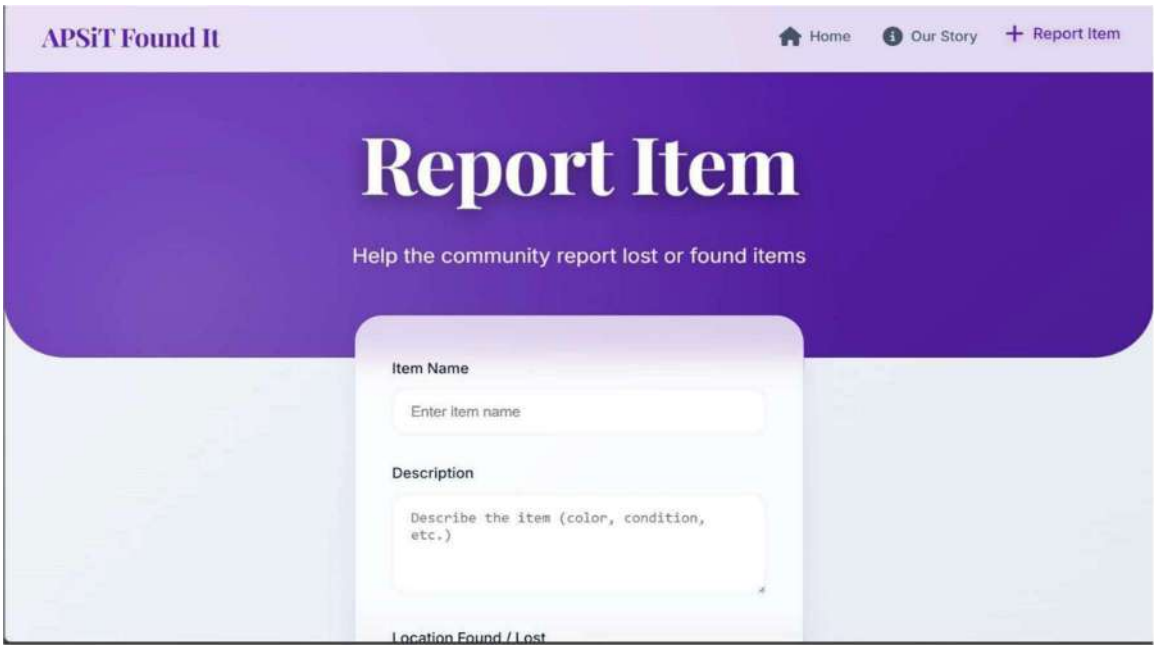


FIGURE 6: Report Item

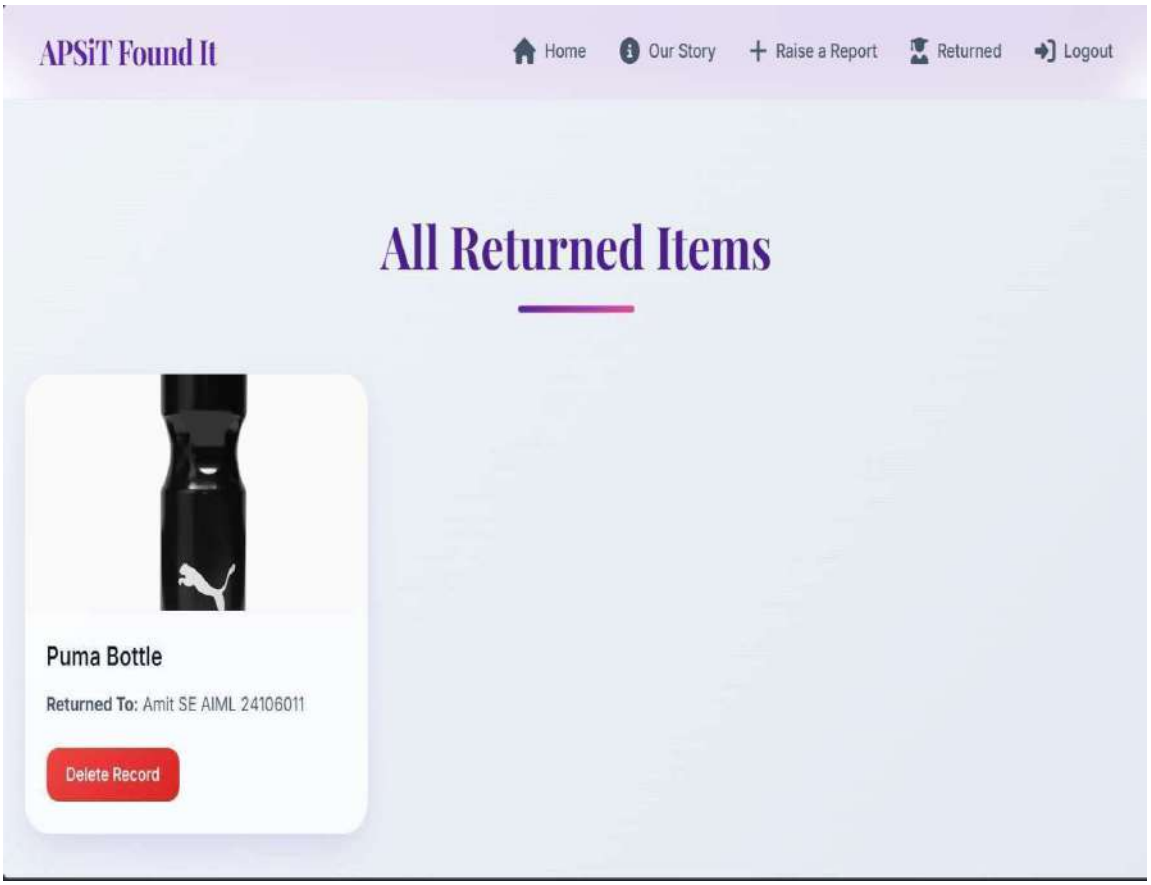


FIGURE 7: **Lost item Page**

Chapter 8

Conclusion and Future Work

8.1 Conclusion

The "Institute Level Lost and Found Portal with Senior-to-Junior Academic Exchange System" successfully addresses the major challenges associated with managing lost items in educational institutions. The system replaces traditional manual methods with a centralized digital platform that improves efficiency, accuracy, and transparency. Users can report lost or found items, search records, and submit claims through a structured workflow, which simplifies the overall process of item recovery.

One of the key achievements of the system is the improvement in recovery rate and processing speed. The use of a searchable database, along with features such as image uploads and keyword-based search, makes it easier for users to identify and locate items. The structured claim verification process ensures that items are returned to their rightful owners, reducing misuse and increasing trust. The system also maintains proper records of all transactions, improving accountability and system monitoring.

Another important contribution is the integration of the Senior-to-Junior academic exchange module. This feature allows students to share or sell academic materials, promoting resource utilization and reducing financial burden. Overall, the system provides a reliable, efficient, and scalable solution that enhances user experience and supports digital transformation in educational institutions.

8.2 Future Work

Although the proposed system effectively addresses current challenges, there are several opportunities for future improvements. One major enhancement is the development of a mobile application, which will provide better accessibility and allow users to interact with the system more conveniently from their smartphones.

Another important improvement is the integration of artificial intelligence for automatic item matching. Image recognition techniques can be used to compare lost and found items, reducing manual effort and improving accuracy.
Additionally, implementing notification systems such as SMS or email alerts can keep users informed about updates related to their items or claims.

The system can also be enhanced by adding an online payment feature within the academic exchange module, making transactions more secure and convenient. Furthermore, cloud deployment can improve scalability and allow the system to be used across multiple institutions. These enhancements will make the system more advanced, efficient, and suitable for real-world applicati

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